GRADE 3 · DEBUGGING

The Wrong Block Inside

Ontario Math C3.2 — read, find, and fix code

Name: .... Date: ....

★ **I can find and fix a wrong block inside a loop.**

Right count, wrong block

Sometimes the repeat number is correct, but a BLOCK inside the loop is wrong. The loop runs the right number of times — it just does the wrong action.

Read what Bit should do, then find the inside block that does not match.

Bit should clap then STOMP — but this is buggy

```
repeat 3
    clap
    clap
```

The repeat 3 is right, but the second block should be STOMP, not clap. Fix the second block.

1 Find the bug

Bit should do clap, stomp — but the loop has clap, clap. Which block is wrong, and what should it be?

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2 Fix the turtle

This should draw a SQUARE: for i in range(4): bit.forward(50); bit.forward(50). But Bit draws a straight line! Which inside block is wrong, and what should it be?

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3 Write the fix

Bit should do jump, spin, but the loop has jump, jump. Write the block that the SECOND one should be: ______

4 Challenge

The square code for i in range(4): bit.forward(50); bit.forward(50) makes Bit go forward 8 times in a line. WHY is it a line and not a square? What block is missing?

Big idea

A loop can repeat the right number of times but still be wrong if a block INSIDE is wrong. Check that every inside block matches what Bit should do.

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Quick facilitation notes & answer key

What this worksheet teaches

Students find a loop where the repeat number is right but a block INSIDE is wrong — in block and shape cases — and describe how the wrong block changes the result (a square becomes a straight line). The method here is SHOULD → DOES → FIX: say what the code SHOULD do, run it to see what it DOES, then FIX the one thing that is wrong. No coding experience needed.

Before you start

No computer needed — the worksheet shows the code and what Bit should do. Optional: a Scratch-style app (for the block loops) or a free Python-turtle playground (for the shapes) to run the code — you only press Run.

How to lead it (about 30 minutes)

1. Show it (you do). The repeat number is right, but an inside block is wrong, so the result is wrong.

2. Do one together. Exercise 1 — the second block should be stomp, not clap.

3. Let them work. Students fix a square drawn as a line (Ex 2–3) and explain why two forwards with no turn make a line (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — the second block is wrong; change clap to stomp. Exercise 2 — the second "move forward" is wrong; change it to "turn right 90".

Exercise 3 — spin. Exercise 4 (challenge) — with two forwards and no turn, Bit just keeps going straight (8 forwards in a line); the turn block is missing.

If students get stuck — and ways to adjust

Watch for: changing the repeat number. Here the count is right; an inside block is wrong.

Make it easier: compare the goal pattern to the blocks inside, one at a time.

Make it harder: ask which inside block makes a square versus a line.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, find what is wrong, fix it, and explain what the change did.

You'll know it worked when

The child keeps the repeat number and fixes the wrong inside block.